Paper 1 - An artificial intelligence model for sand and dust storm forecast driven by AI weather forecasts

Inputs -

1. **Dust Concentration (at time t)**
   - Surface
   - 950 hPa
   - 850 hPa
   - 500 hPa
2. **Wind Fields (at time t and t+Δt)**
   - Surface
   - 1000 hPa
   - 950 hPa
   - 850 hPa
   - 700 hPa
   - 500 hPa
3. **Temperature (at time t)**
   - Surface
   - 1000 hPa
   - 950 hPa
   - 850 hPa
   - 700 hPa
   - 500 hPa
4. **Dust Advection** (wind-driven dust transport)
5. **Dust Emission Flux**
6. **Vegetation Cover (NDVI)**
7. **Relative Terrain Height**
8. **Land Use Information**
9. **Surface Roughness Density**
10. **15 × 15 Spatial Neighborhood Grid**

    Sources -

- **Dust concentration → CAMx simulation**
- **Wind & Temperature → WRF simulation**
- **Dust advection → Computed from wind + dust concentration**
- **Dust emission → Physics-based dust emission scheme**
- **Vegetation cover → MODIS NDVI (MOD13Q1)**
- **Soil texture → SoilGrids250m**
- **Grain size → Wei et al. & Gong et al.**
- **Relative terrain height → Terrain data from model domain**
- **Land use → Static geographic data**

- **Surface roughness density → High-resolution roughness dataset**

Outputs -

Predicted Dust Concentration at the next time step (t + Δt) at:

- Surface
- 950 hPa
- 850 hPa
- 500 hPa

Paper 2 - Using Hybrid Deep Learning Models to Predict Dust Storm Pathways with Enhanced Accuracy

The paper uses **MERRA-2 (NASA)** data as the primary source. The inputs are:

1. **Aerosol Optical Depth (AOD)** images (dust particle information)
2. **Surface Wind Speed**
3. **Surface Wind Direction**
4. **Relative Humidity**
5. **Surface Air Temperature**
6. **Surface Skin Temperature**
7. **Wind direction at 10 m above ground**
8. **Wind direction at 50 m above ground**

The model predicts the **dust storm transport pathway (movement and shape)** for the next **24 hours**.

Outputs - Predicted dust storm pathway/location for the next 24 hours (6h, 12h, 18h, 24h forecasts).

Paper 3 - Dust storm detection of a convolutional neural network and a physical algorithm based on FY-4A satellite data

Input - The CNN model takes **8 spectral bands from the FY-4A AGRI satellite** as input

Output - Binary **dust storm mask** classifying each pixel as **Dust (1)** or **Non-Dust (0)** and identifying the spatial extent of dust storms